

# MP5 MiniClaw

From Wrong Requirements to Honest Design

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# The Brief

What Jordan asked us to build

## Jordan Chen Design Brief — MP1

Compete with Hiwonder's BigClaw at RobotExpo 2026

Envelope:  $92 \times 46 \times 55$  mm

Jaw opening:  $\geq 40$  mm

Thumb-wheel turns: 2–3 from open to closed

Grip force: 5–8 N

Material: FDM PLA (Prusa MK4S)

Part count:  $< 15$  parts

Cost:  $< \$3$ /unit at 500-unit scale

"The BigClaw outsells us 3-to-1. Your MiniClaw needs to match its jaw range in half the volume."

— Jordan Chen, Engineering Manager

# Where We Ended Up

Final MiniClaw design vs. MP1 targets

| Specification       | MP1 Target      | Final Design    | Status       |
|---------------------|-----------------|-----------------|--------------|
| Jaw opening         | $\geq 40$ mm    | 40.13 mm        | Met          |
| Envelope            | 92 × 46 × 55 mm | 92 × 46 × 55 mm | Met          |
| Thumb-wheel turns   | 2–3             | 0.28            | Renegotiated |
| Gear stress SF      | $\geq 2.0$      | 2.14            | Met          |
| Part count          | < 15            | 25              | Over target  |
| Trans. angle margin | > 0° above 40°  | 21.3° margin    | Met          |

# The Renegotiation Story

GOAL & DIRECTION

## Why we accepted 0.28 turns instead of 2–3

### Architecture B (compound spur)

$N = 16 \rightarrow 1.24$  thumb-wheel turns

Center distance: 126 mm total

Exceeds 92 mm housing by 34 mm

Part count: 29 (5 gears)

Packaging: DOES NOT FIT

### Architecture A (single spur pair)

$N = 1 \rightarrow 0.28$  thumb-wheel turns

Center distance: 20 mm

Fits easily in 92 mm housing

Part count: 25 (2 gears)

Packaging: SOLVED

# When Context Was Missing

One document upload corrected the entire design basis

| Parameter          | AI Without Context | After MP1 Brief Upload |
|--------------------|--------------------|------------------------|
| Jaw opening target | 40 mm              | 25 mm                  |
| Input range        | 0°–46°             | 0°–28°                 |
| Link lengths       | L1=40, L2=20       | L1=12, L2=26           |
| Displacement       | 20.3 mm/side       | 12.5 mm/side           |

## Lesson learned:

The AI will confidently build on wrong assumptions if you don't provide the requirements document. Context management isn't optional.

Also missing: Haben's repo was private (403) until made public. Yoel's MP4 Part A was never submitted.

| Tool                    | Used For                                                            |
|-------------------------|---------------------------------------------------------------------|
| Devin (Cognition AI)    | Part A notebooks, Part B templates, plots, stress analysis, 12+ PRs |
| GitHub Copilot (agent)  | Haben's Part A, David's MP3 gear CAD                                |
| MiniClaw Gear DFM Skill | Yoel's MCP RAG server + ACME corpus (ACME-ENG-001, MFG-002)         |
| matplotlib / numpy      | Comparison plots, linkage simulation, drive-train sketch            |
| GitHub                  | Version control, code review, team integration                      |

**Live demo coming up: animated four-bar simulation + `compute_transmission_angle()` on perturbed geometry**

# The Pin Clearance Loop

**Human × AI = something neither would have produced alone**

David asked: "What bore diameter gives a reliable sliding fit after FDM tolerance?"

| Bore Ø (mm) | Designed | FDM Offset | Post-FDM | Verdict      |
|-------------|----------|------------|----------|--------------|
| 3.20        | 0.20 mm  | −0.20 mm   | 0.00 mm  | INTERFERENCE |
| 3.30        | 0.30 mm  | −0.20 mm   | 0.10 mm  | Tight        |
| 3.40        | 0.40 mm  | −0.20 mm   | 0.20 mm  | RELIABLE FIT |

Human: FDM tolerance data from MP1  
AI: systematic clearance stack calculation

**Result: bore increased to Ø3.40 mm across  
all 10 joints. No reaming needed.**

# The Wrong-AI Moment

EVALUATION & TRUST

## What the AI claimed

Designed David's linkage targeting:

- 40 mm jaw opening
- 0°–46° input range
- L1=40, L2=20 link lengths

Complete design with position equations, transmission angle analysis, motion plots, and hand calculations.

Everything checked out internally.

## What was actually true

David's MP1 brief specified:

- 25 mm jaw opening
- 0°–28° input range
- L1=12, L2=26 link lengths

The AI had never seen the MP1 requirements document. It guessed the specs and built a plausible-looking but wrong design.

## How we caught it

David compared the AI's output against his original design brief.

The numbers didn't match.

One document upload fixed everything.

Mechanism: the human knew the requirements. The AI didn't.

**Context management isn't optional — it's the difference between a correct design and a plausible-looking wrong one.**



# Per-Subsystem Trust Assessment

EVALUATION & TRUST

| Subsystem      | Flag       | Print Tomorrow? | Key Issue                                        |
|----------------|------------|-----------------|--------------------------------------------------|
| 1. Linkage     | Needs work | YES             | Strongest analytically (3-path verified)         |
| 2. Drive Train | Needs work | YES             | SF = 2.14, but 0.28 turns untested ergonomically |
| 3. Jaw Arms    | Unknown    | NO              | No CAD exists                                    |
| 4. Housing     | Unknown    | NO              | No CAD exists                                    |
| 5. Input Wheel | Unknown    | NO              | No CAD exists                                    |
| 6. Pins        | Needs work | YES             | Ø3.40 bore designed; test one joint first        |

**Bottom line: Not ready to print a complete gripper. Three subsystems are analytically solid; three have no CAD. This is an honest assessment.**

# What We'd Do Differently

## 1. Shared team repo from Day 1

Cross-repo integration (cloning Haben's private repo, cloning Yoel's repo, extracting parameters) added friction. A single shared workspace from the start would have eliminated the 403 errors, the manual data extraction, and the integration PRs.

## 2. Build the MCP tool call earlier

Having `compute_transmission_angle()` live during linkage selection (Loop 1) would have made the comparison interactive instead of static. The tool we demo today should have existed in Week 4.

# Live Demo

Yoel Tesfatsion

## Part 1: Dynamic Simulation

Haben's four-bar linkage sweeping  $-50^\circ$  to  $+50^\circ$

Jaw opening from 0 to 40 mm and back

Transmission angle staying in workable band

Real-time data display

## Part 2: Live Tool Call

`compute_transmission_angle(L3=32)`

vs. our design ( $L3=29$ )

Shows angle drops with 3 mm change

Tool catches bad geometry instantly

```
$ Terminal
```

```
$ python3 linkage_simulation.py
```

```
$ python3 transmission_angle_check.py
```

```
--- Haben's design ---
```

```
μ range: 61.3° - 91.3°
```

```
Margin: 21.3° above 40°
```

```
--- Perturbed: L3=32 ---
```

```
μ range: 56.0° - 84.1°
```

```
Margin: 16.0° above 40°
```

# Questions?

| Pillar              | Answerer        |
|---------------------|-----------------|
| Goal & Direction    | David Ricciotti |
| Context Management  | Haben Berhe     |
| Tools & Integration | Yoel Tesfatsion |
| Centaur Engineering | Haben Berhe     |
| Evaluation & Trust  | David Ricciotti |

**Thank you!**